Simon Hirst

SOFTWARE ENGINEER - REAL-TIME / PRODUCT ENGINEERING

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PROFILE

Software engineer with seven years of commercial experience across TypeScript/Node.js, React and C#/.NET. Recent work includes real-time media systems with WebRTC, WebSockets, failover and automated testing; earlier roles span customer-facing applications, APIs, Azure integrations and production reliability. Strong in debugging, testability and incremental improvement of existing systems.

CORE SKILLS

Languages	TypeScript/JavaScript, C#, Go, Java, SQL
Product / web	React, Next.js, Node.js, Express, REST APIs
Real-time	WebSockets, WebRTC, Server-Sent Events, RabbitMQ, Redis
Platform	Azure, Docker, Kubernetes, OpenShift, GitHub Actions
Operations	Prometheus, Grafana, structured logging, tracing
Testing	xUnit, Jest, Playwright

PROFESSIONAL EXPERIENCE

Freelance Software Engineer - Streaming infrastructure, Remote Apr 2026 - present

- Built a Node.js live-streaming stack using WebRTC, obs-websocket, MediaMTX and SRTLA, with automatic failover, real-time telemetry/audio and 146 automated tests.

Cirium - Software Engineer, Southampton (hybrid) Nov 2025 - Apr 2026

- Shipped a production fix to JFK flight-cancellation handling affecting over 100 flights per day, validating the result with SQL in staging and production.
- Diagnosed late and out-of-order flight messages across three connectors using logs, database queries and monitoring.

Freelance Software Engineer - Remote Aug 2024 - Nov 2025

- Built Node.js/.NET web applications with Next.js, webhooks, automated tests, CI and telemetry; also developed a TypeScript/WebGL2 audio visualisation engine.

IBM - Software Engineer, Remote Jul 2021 - Jul 2024

- Maintained IBM MQ deployment configuration, Helm charts and manifests across Kubernetes and OpenShift for over ten supported environments.
- Wrote Go tooling to validate cluster prerequisites before installation, reducing failed installations caused by environment misconfiguration.
- Added Prometheus metrics and Grafana dashboards used by four teams, and led investigations spanning high availability, storage, networking, upgrades and disaster recovery.

Dootrix - Software Engineer, Remote Mar 2020 - Jul 2021

- Built .NET and React onboarding and account-management flows including email verification, password reset and multi-factor authentication.
- Developed REST APIs and Azure Functions for CRM and queue-based integrations, with xUnit, Jest and Playwright coverage plus logging and tracing.

Zupa - Software Engineer, Southampton Sep 2019 - Mar 2020

- Cut a nightly reconciliation and notification job from over five hours to thirty minutes by optimising SQL queries and background job scheduling, and added operational alerting around it.
- Developed ASP.NET Core checkout and Stripe payment flows with validation, idempotency and safe retry handling.

EDUCATION

BSc Computer Science (2:1) - University of Liverpool